🚚 On-Road Vehicle Breakdown Assistance Finder

1. Introduction

The **On-Road Vehicle Breakdown Assistance Finder** is a mobile-friendly web application designed to help vehicle users receive quick assistance during breakdown situations. The platform connects **Users**, **Verified Mechanics**, and an **Admin** in a secure ecosystem with OTP authentication, risk analysis, and real-time tracking.

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2. Objectives

- Provide instant roadside assistance
- Prevent fake users and mechanics using OTP & admin verification
- Estimate breakdown risk and service cost using KNN
- Offer real-time GPS tracking for mechanics and cabs
- Build a mobile-first web app extendable to a future mobile app

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3. Technology Stack

Frontend

- HTML5
- CSS3 (Responsive / Mobile-First)
- JavaScript

Backend

- Node.js
- Express.js

Database

- MongoDB (NoSQL)

External Services

- SMS OTP Gateway (Twilio / Firebase)
- Google Maps API (GPS & location)

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4. System Roles

4.1 User

- Register & Login using OTP
- Add and manage vehicles
- Request breakdown assistance
- View mechanic and cab live tracking

4.2 Mechanic

- Register using OTP
- Submit documents for verification
- Await admin approval
- Accept service requests
- Share live GPS location

4.3 Admin

- Approve / reject mechanics
- Monitor users, mechanics, and requests
- View analytics and system statistics

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5. Authentication & OTP Verification

OTP Flow (User & Mechanic)

1. Enter mobile number
2. Receive 6-digit OTP
3. Verify OTP (expires in 2–5 minutes)
4. Account activated

Security Features

- OTP retry limit
- OTP expiry
- Hashed passwords

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6. User Module Workflow

New User

- Sign up → OTP verification
- Add vehicle details

- Directly access assistance finder

Existing User

- Login → OTP verification
- Manage multiple vehicles
- Request assistance anytime

Breakdown Assistance Finder

- Select vehicle
- Auto-detect location (GPS)
- Select issue type
- Get risk analysis

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7. KNN-Based Breakdown Analysis

Input Parameters

- Vehicle type
- Problem type
- Previous breakdown history
- Distance traveled

Output

- Breakdown factor
- Risk level (Low / Medium / High)
- Estimated service cost

Action Logic

- Low: Self-help tips
- Medium: Suggest nearby mechanic
- High: Immediate mechanic + towing

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8. Mechanic Module Workflow

Registration

- OTP verification
- Upload documents
- Status: Pending Approval

After Admin Approval

- View nearby service requests
- Accept or reject requests
- Share live GPS tracking
- Update service status

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9. Cab Service Module

- Triggered if repair takes > 1 day
- Book cab via application
- Live cab tracking for user

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10. Admin Module

Admin Dashboard

- Total users
- Verified mechanics
- Pending approvals
- Requests per user

Admin Controls

- Mechanic verification
- System monitoring

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11. Database Design (MongoDB)

Users Collection

```
{
  name: String,
  mobile: String,
  otpVerified: Boolean,
  vehicles: Array,
  requests: Array
}
```

Mechanics Collection

```
{
  name: String,
  mobile: String,
```

```
  otpVerified: Boolean,
  adminApproved: Boolean,
  services: Array,
  location: Object
}
```

Requests Collection

```
{
  userId: ObjectId,
  mechanicId: ObjectId,
  problemType: String,
  riskFactor: String,
  status: String
}
```

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12. API Modules (Backend)

- /auth/register
- /auth/login
- /auth/verify-otp
- /user/add-vehicle
- /request/create
- /mechanic/approve
- /tracking/live

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13. Frontend Pages

- Login Dashboard (User / Mechanic / Admin)
- OTP Verification Page
- User Dashboard
- Mechanic Dashboard
- Admin Dashboard
- Live Tracking Page

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14. Security Measures

- Role-based access control
- OTP authentication
- Admin verification
- Secure password hashing

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15. Advantages

- Fast roadside help
- Trusted mechanics only
- Real-time tracking
- Cost transparency
- Mobile-friendly design

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16. Future Enhancements

- Android & iOS apps
- Online payment gateway
- AI-based diagnostics
- Emergency SOS button
- Push notifications

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17. Conclusion

The **On-Road Vehicle Breakdown Assistance Finder** is a complete, secure, and scalable solution for roadside emergencies. With OTP verification, admin-approved mechanics, intelligent risk prediction, and real-time tracking, the system ensures safety, reliability, and convenience for users. The project is future-ready for mobile app deployment.